PhysCC: A Physics-as-a-Compiler Framework for HPC Simulations

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February 2026

1 Introduction

Modern scientific computing often suffers from a "semantic gap" between high-level physical laws and low-level hardware optimizations. This project implements *PhysCC*, a domain-specific compiler designed to bridge this gap by treating physical equations as source code and hardware-specific instructions as the target machine.

2 Methodology: Parallel Pipeline Architecture

The core methodology of PhysCC is grounded in the side-by-side mapping of classical computer science compiler phases to physical semantics[cite: 2, 3].

Table 1: Parallelism between CS Compilers and Physics-as-a-Compiler

Compiler Phase	**CS Semantic**	**Physics Semantic**
Source Code	Textual Logic	Physical Law (PDE/QM)
Lexical Analysis	Tokens	Symbolic Notation (d, V, ∇^2)
AST	Syntax Tree	Operator Tree
Type Checking	Data Invariants	Physical Invariants/Dimensions
IR	Three-Address Code	Discrete Stencils/Matrices
Optimization	Resource Tuning	Stability & Error Minimization
Lowering	Binary Mapping	Discretization (FD/FEM)
Backend Codegen	Machine Code	Solver/Circuit Generation

3 The Compilation Bridge

The key to this approach is using the compilation process to bridge abstraction to hardware. By maintaining physical semantics at every step, the compiler ensures that the generated code is not only fast but mathematically correct and stable[cite: 5].

4 Conclusion

PhysCC demonstrates that treating Linear Algebra and Physics as a first-class programming language allows for the generation of highly optimized code for CPU (AVX2), GPU (SYCL/CUDA), and Quantum hardware without sacrificing the readability of the underlying physics.